Technical Description

Exhaust gas heat exchanger – Technical overview

MS61023/00E

SLU Name	Sales Name
GG08V4000A1	MTU 8V4000 GS
GG08V4000A2	MTU 8V4000 GS
GG12V4000A1	MTU 12V4000 GS
GG12V4000A2	MTU 12V4000 GS
GG16V4000A1	MTU 16V4000 GS
GG16V4000A2	MTU 16V4000 GS
GG20V4000A1	MTU 20V4000 GS
GG20V4000A2	MTU 20V4000 GS

The original document was created in the German language.

All of the information presented in this publication was current and up-to-date at the time of publishing. MTU Onsite Energy GmbH reserves the right to make changes to, delete, or supplement the provided information and data as needed.

Preface

These brief instructions serve as an aid and provide information on the installation and purchasing of MTU components for planning or design.

The brief instructions apply to the respective MTU components.

The objective of these brief instructions is to provide correct component information.

The brief instructions do not release persons responsible for the system from their responsibility for correct design and inspection.

This document serves as an overview of the dimensions and the appearance of the specified options with the most important technical data.

The specifications display the corresponding components so that they can be identified on the collective drawing included here, e.g. (AWT4000_BG_1).

The specified measures and dimensions are binding. The necessary installation space and accessibility for operation and maintenance must be guaranteed by the customer. The installation conditions must be observed.

The relevant documents are defined in the specifications related to the order concerned.

Performing prescribed maintenance tasks conscientiously and on schedule positively influences operational safety, reliability and service life. Bear in mind accessibility for operation, maintenance and repair work when planning and installing the system.

Definition of Info box

This symbol with text box is used solely as an Info box for the reader.
It does not contain any type of references to situations where damage to material or personal injury are possible!
It must only contain information and tips related to the product.

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Table of Contents

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1 Exhaust gas heat exchanger

1.1 General

1.1.1 Components

Assembly consists of:

- Exhaust gas heat exchanger basic body
- Safety fittings assembly
- Heating water connection set
- Exhaust gas connection set

The components are delivered loosely and have to be assembled by the customer on site.

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1.2 Basic body

1.2.1 General information

- Exhaust gas heat exchanger as per Pressure Equipment Directive
- No ASME certification

Design of exhaust gas heat exchanger	
Design	Tube bundle heat exchanger
Flow guide	Counterflow
Design	horizontal, including feet

Materials	
Pipe plates	1.4571
Tubes	1.4571
Inlet and outlet chamber	1.4571
Housing	Steel
Exterior parts and miscell.	Steel

Temperatures	
Exhaust gas temperature, outlet	120 °C
Heating water, inlet	80 °C
Heating water temperature, outlet	90 °C
Maximum continuous operating temperature, exhaust side	550 °C
Maximum short-term operating temperature, exhaust side	600 °C

Pressure	
Minimum operating pressure, water system	10 bar (overpressure)

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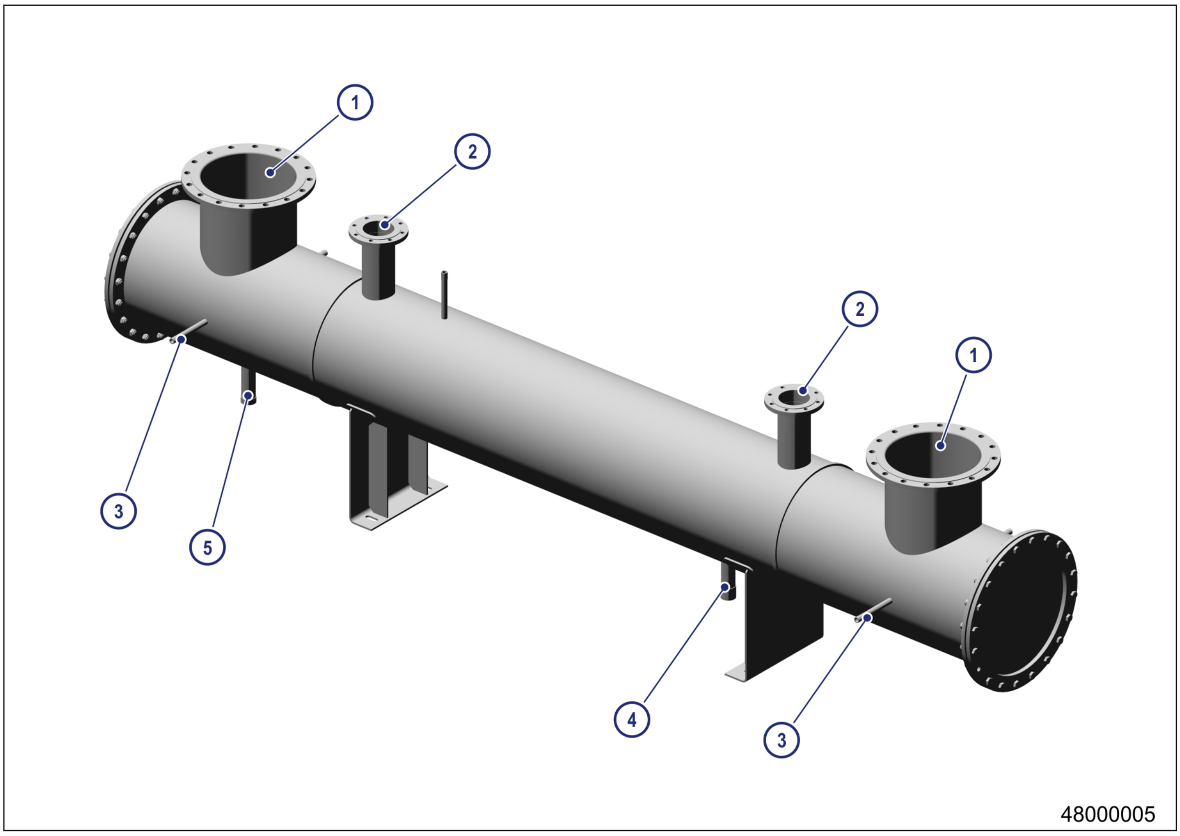

Figure 1: Exhaust gas heat exchanger positions

1 Exhaust gas outlet / inlet
2 Heating water inlet / outlet
3 Pressure / temperature measurement connection DN15
4 Drain DN50
5 Condensate trap DN50

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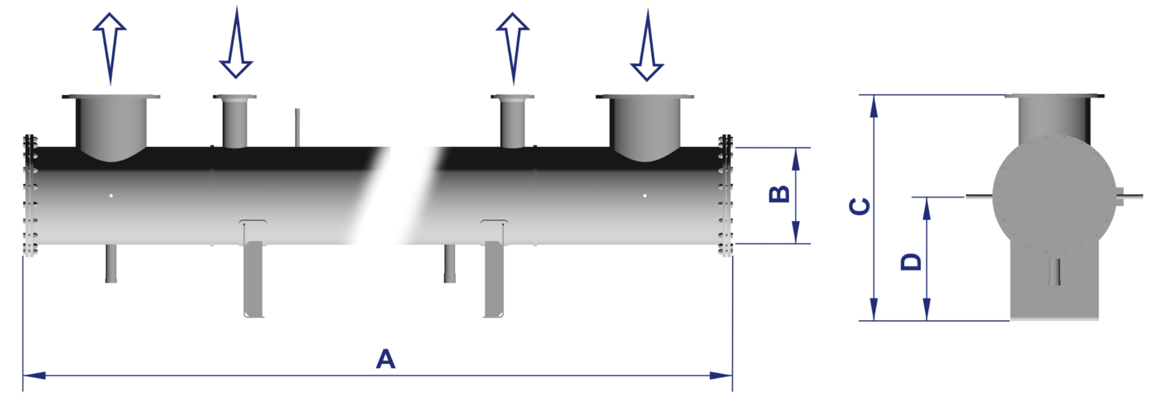

Figure 2: Exhaust gas heat exchanger dimensions

Exhaust gas heat exchanger units										
Type Exhaust gas heat exchanger	Length [mm]	Diameter [mm]	Top edge Exhaust gas heat exchanger [mm]	Center Exhaust gas heat exchanger [mm]	Flange Exhaust [DN/PN]	Flange Heating water [DN/PN]	Weight, empty Exhaust gas heat exchanger [kg]	Water content [l]	Pressure loss Exhaust side [mbar]	Pressure loss Heating water side [mbar]
	A	B	C	D						
AWT4000_BG_1	4600	508	1228	674	300/10	100/16	890	245	12–15	68
AWT4000_BG_2	4820	550	1270	695	400/10	125/16	1040	280	15–17	61
AWT4000_BG_3	4570	650	1370	745	400/10	125/16	1270	345	10–14	63
AWT4000_BG_4	4930	650	1370	745	500/10	125/16	1365	365	14–16	77
AWT4000_BG_5	4885	750	1470	795	500/10	150/16	1580	550	11–15	83
AWT4000_BG_6	4500	700	1420	770	500/10	150/16	1480	445	16–17	93

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Exhaust gas heat exchanger units										
AWT4000_BG_7	4400	750	1470	795	600/10	150/16	1690	460	11–16	85
AWT4000_BG_8	3500	550	1270	695	300/10	100/16	960	245	15–16	62

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